

Keefe Ker Tan

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EDUCATION

Singapore Institute of Technology Bachelor of Engineering with Honours in Robotics Systems (Expected Jan 2027)	Sep 2024 – Present
Republic Polytechnic Diploma in Engineering System and Management	Apr 2021 – Apr 2024
Institute of Technical Education Higher Nitec in Information Technology	Apr 2016 – Apr 2018

EXPERIENCE

Singapore Armed Forces Infantry Trooper (Full-time National Service) - Appointed Section 2IC (Second-in-Command) for performance and leadership potential - Awarded IPPT Gold and Marksmanship Award for physical excellence and combat proficiency - Received personal Commander's appraisal recognising outstanding service, quick learning, and strong work ethic	Oct 2018 – Aug 2020
Timberland / Under Armour Retail Sales Associate - Delivered strong customer service and product recommendations tailored to customer needs - Consistently met or exceeded sales targets and contributed to a positive team environment	Nov 2014 – Aug 2018

PROJECTS

Warehouse Utilization Monitoring System Industry-Partnered Project Team of 4 Singapore Institute of Technology - Built a computer vision pipeline using YOLOv8 to detect pallets and monitor warehouse staging zone utilization, meeting the project's spatial accuracy requirements - Contributed to system design and dashboard development; prototype selected for potential further development with the industry partner	Jan 2026 – Mar 2026
Autonomous Maze Navigation Individual Project Singapore Institute of Technology Developed a ROS2 navigation node for autonomous maze exit in Gazebo simulation	Sep 2025 – Dec 2025
Autonomous Navigation with LIMO Robot Team of 6 Singapore Institute of Technology - Worked with a team of 6 to design and build a themed physical arena (Terminal 4, Changi Airport) and develop ROS1 navigation scripts for goal-based waypoint navigation on the AgileX LIMO robot - Contributed to both the physical arena construction and navigation coding, delivering a working system before the final assessment	May 2025 – Jul 2025

SKILLS

Programming & Tools: C, C++, Python, HTML, CSS, ROS1, ROS2, Git, Linux (Ubuntu)
Software & Frameworks: RViz, Gazebo, Nav2, YOLOv8, OpenCV, Flask, ActionLib, move_base, SLAM, EKF
Hardware: AgileX LIMO, Orbbec Dabai Camera, STM32 Nucleo-64, Pixy2, Mecanum Wheels, Sensor Integration
Leadership Skills: Team Collaboration, Communication, Adaptability, Problem-Solving, Decision-Making
Languages: English, Mandarin, Hokkien